**Sample Midterm 1, MATH 4211/6211 Optimization**

**NAME:**________________________**ID#:** __________________ **Section: 4211/6211**

**Show all your work. Answers without solution steps will not receive credits.**

1. (4 points) Show that $\{(\boldsymbol{x}, t) \in \mathbb{R}^{n+1} : \|\boldsymbol{x}\| \le t, t \ge 0\}$ is a convex set in $\mathbb{R}^{n+1}$ using the definition.

   **Solution.** Denote the set by $C$. For any $(\boldsymbol{x}, t), (\tilde{\boldsymbol{x}}, \tilde{t}) \in C$, there are $\|\boldsymbol{x}\| \le t$ and $\|\tilde{\boldsymbol{x}}\| \le \tilde{t}$. Hence for any $\theta \in [0, 1]$, there is

$$\|\theta\boldsymbol{x} + (1-\theta)\tilde{\boldsymbol{x}}\| \le \theta\|\boldsymbol{x}\| + (1-\theta)\|\tilde{\boldsymbol{x}}\| \le \theta t + (1-\theta)\tilde{t},$$

   which implies that $\theta(\boldsymbol{x}, t) + (1-\theta)(\tilde{\boldsymbol{x}}, \tilde{t}) = (\theta\boldsymbol{x} + (1-\theta)\tilde{\boldsymbol{x}}), \theta t + (1-\theta)\tilde{t}) \in C$. Therefore, $C$ is a convex set.

2. (4 points) Suppose $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_m \in \mathbb{R}^n$ and $b_1, \ldots, b_m \in \mathbb{R}$ are given. Define $f : \mathbb{R}^n \to \mathbb{R}$ as $f(\boldsymbol{x}) = -\log(-\log(\sum_{j=1}^m e^{\boldsymbol{a}_i^T \boldsymbol{x} + b_i}))$ for $\boldsymbol{x} \in \mathbb{R}^n$. Is $f$ a convex function, and why? You can use the fact that $\log(\sum_{i=1}^n e^{x_i})$ is convex in $\boldsymbol{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$.

**Solution.** Denote $A = [\boldsymbol{a}_1, \ldots, \boldsymbol{a}_m]^T \in \mathbb{R}^{m \times n}$ and $\boldsymbol{b} = (b_1, \ldots, b_m)^T \in \mathbb{R}^m$. Also denote $g(\boldsymbol{y}) = -\log(\sum_{j=1}^m e^{y_j})$ for all $\boldsymbol{y} \in \mathbb{R}^m$, and $h(x) = -\log(z)$ for any $z > 0$. Then $f(x) = h(g(A\boldsymbol{x} + \boldsymbol{b}))$. Note that $g$ is concave (since $-g$ is convex), we know $g(A\boldsymbol{x} + \boldsymbol{b})$ is concave since it is a composition of a concave function and an affine function. Also $h$ is convex and decreasing, so $h(g(A\boldsymbol{x} + \boldsymbol{b}))$ is convex in $\boldsymbol{x}$. Hence $f$ is convex in $\boldsymbol{x}$.

3. (4 points) Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(\boldsymbol{x}) = \boldsymbol{x}^T \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} \boldsymbol{x} + \boldsymbol{x}^T \begin{bmatrix} 3 \\ 5 \end{bmatrix} + 6$$

(a) Find the gradient and the Hessian of $f$ at $\boldsymbol{x} = [1, 1]^T$.

(b) Find the directional derivative of $f$ at $\boldsymbol{x} = [1, 1]^T$ with respect to a unit vector in the direction of maximal rate of increase.

**Solution.**

(a) We first compute the first and second order partial derivatives:

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= 2x_1 + 6x_2 + 3 \\ \frac{\partial f}{\partial x_2} &= 6x_1 + 14x_2 + 5 \\ \frac{\partial^2 f}{\partial x_1^2} &= 2 \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} &= \frac{\partial^2 f}{\partial x_1 \partial x_2} = 6 \\ \frac{\partial^2 f}{\partial x_2^2} &= 14. \end{aligned}$$

Therefore we obtain the gradient and Hessian of $f$ as

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} 2x_1 + 6x_2 + 3 \\ 6x_1 + 14x_2 + 5 \end{bmatrix} \quad \text{and} \quad \nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 2 & 6 \\ 6 & 14 \end{bmatrix}$$

and any given point $\boldsymbol{x} \in \mathbb{R}^2$. Plugging in $\boldsymbol{x} = [1, 1]^T$, we obtain $\nabla f([1, 1]^T) = [11, 25]$ and $\nabla^2 f([1, 1]^T) = \begin{bmatrix} 2 & 6 \\ 6 & 14 \end{bmatrix}$.

(b) From calculus we know that the direction of maximal rate of increase of $f$ at $\boldsymbol{x}$ is given by $\boldsymbol{d} = \frac{\nabla f(\boldsymbol{x})}{\|\nabla f(\boldsymbol{x})\|}$. Therefore we know the directional derivative is

$$\nabla f(\boldsymbol{x})^T \boldsymbol{d} = \nabla f(\boldsymbol{x})^T \frac{\nabla f(\boldsymbol{x})}{\|\nabla f(\boldsymbol{x})\|} = \frac{\|\nabla f(\boldsymbol{x})\|^2}{\|\nabla f(\boldsymbol{x})\|} = \|\nabla f(\boldsymbol{x})\|$$

Plugging in $\boldsymbol{x} = [1, 1]^T$, we obtain $\|\nabla f([0, 1]^T) = \sqrt{11^2 + 25^2} = \sqrt{746}$.

4. (4 points) Define $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(\boldsymbol{x}) = x_1 x_2$ for $\boldsymbol{x} = [x_1, x_2]^T \in \mathbb{R}^2$. Find the gradient and the Hessian of $f$. Is $f$ convex and why?

**Solution.** It is straightforward to calculate the Hessian of $f$ (details omitted here) which is

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

which is not positive semi-definite over $\mathbb{R}^2$. Hence $f$ is not convex.

5. (4 points) Suppose that $\boldsymbol{x}^*$ is a local minimizer of $f$ over the set $\Omega$ where $\Omega \subset \Omega' \subset \mathbb{R}^n$. Show that if $\boldsymbol{x}^*$ is an interior point of $\Omega$ then $\boldsymbol{x}^*$ is a local minimizer of $f$ over $\Omega'$. Show that this cannot be claimed if $\boldsymbol{x}^*$ is not an interior point of $\Omega$.

   **Solution.** Since $\boldsymbol{x}^*$ is a local minimizer of $f$ over $\Omega$, there exists $\delta_1 > 0$ such that $f(\boldsymbol{x}^*) \leq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in \Omega \cap B(\boldsymbol{x}; \delta_1) = \{\boldsymbol{x} : \boldsymbol{x} \in \Omega, \|\boldsymbol{x} - \boldsymbol{x}^*\| \leq \delta_1\}$. Since $\boldsymbol{x}^*$ is an interior point of $\Omega$, then there exists $\delta_2 > 0$ such that $B(\boldsymbol{x}; \delta_2) \subset \Omega$. Choosing $\delta := \min(\delta_1, \delta_2)$, we know $f(\boldsymbol{x}^*) \leq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in B(\boldsymbol{x}^*, \delta) \subset \Omega \subset \Omega'$, which means that $\boldsymbol{x}^*$ is a local minimizer of $f$ over $\Omega'$ as well.

   Consider the example where $f(x) = x$, $\Omega = [0, \infty)$, and $\Omega' = \mathbb{R}$. Then $x^* = 0$ is a local (and also global) minimizer of $f$ over $\Omega$, but $x^*$ is not an interior point of $\Omega$, and it is also not a local minimizer of $f$ over $\Omega'$.